

abc

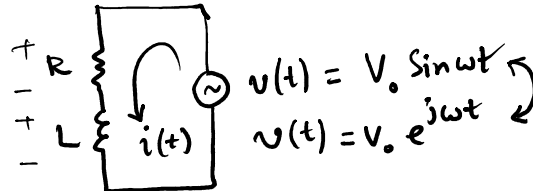
$$\phi(t) = A e^{j\omega t}$$

$$v(t) = V_m e^{j(\omega t + \phi)}$$

$$I = \frac{V e^{j\omega t}}{|Z| e^{j\theta}} = \frac{V}{|Z|} e^{j(\omega t - \theta)}$$

$$Z = |Z| e^{j\theta}$$

R-L circuit



$$\text{Imj}(V_m e^{j\omega t})$$

$$\phi(t) = \text{Amplitude} \cdot e^{j\omega t} \rightarrow \text{time dependence}$$

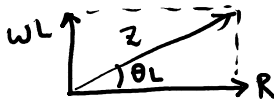
$$v(t) = R i(t) + L \frac{di}{dt}$$

$$V = IR + j\omega L \cdot I$$

$$X_L = j\omega L$$

$$i(t) = I e^{j\omega t}$$

$$v(t) = V e^{j\omega t}$$



$$I = \frac{V}{|Z|} e^{-j\theta_L}$$

$$i(t) = \frac{V}{|Z|} e^{-j\theta_L} \cdot e^{j\omega t}$$

$$V e^{j\omega t} = I \cdot R e^{j\omega t} + L j\omega I e^{j\omega t}$$

$$V = I(R + j\omega L)$$

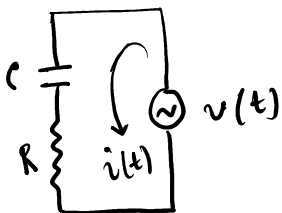
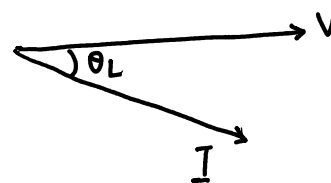
$$I = \frac{V}{R + j\omega L} = \frac{V}{Z} = \frac{V}{|Z| e^{j\theta_L}} = \frac{V}{|Z|} e^{-j\theta_L}$$

$$|Z| = \sqrt{R^2 + \omega^2 L^2} \quad \theta_L = \tan^{-1}(\omega L / R)$$

$$i(t) = \frac{V}{|Z|} e^{j(\omega t - \theta_L)}$$

$$v(t) = V e^{j\omega t}$$

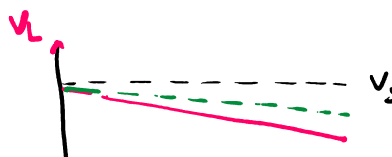
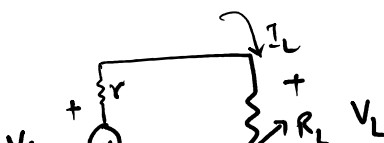
$$\theta_L = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

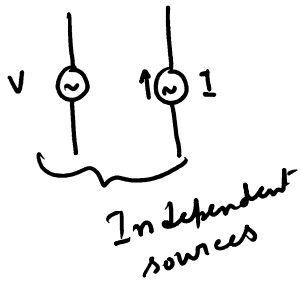
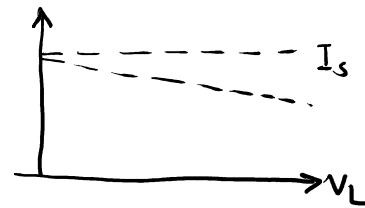
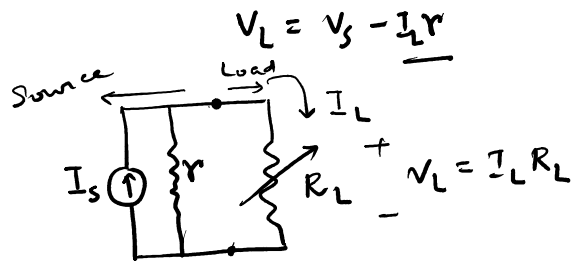
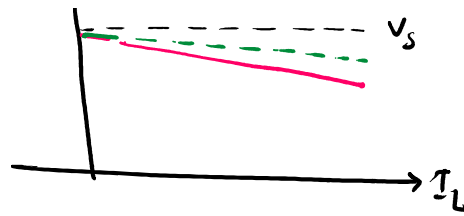
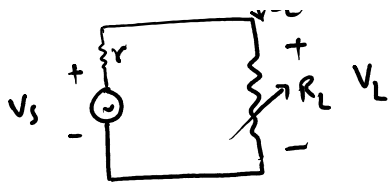


$$v(t) = i(t)R + \frac{1}{C} \int i(t) dt$$

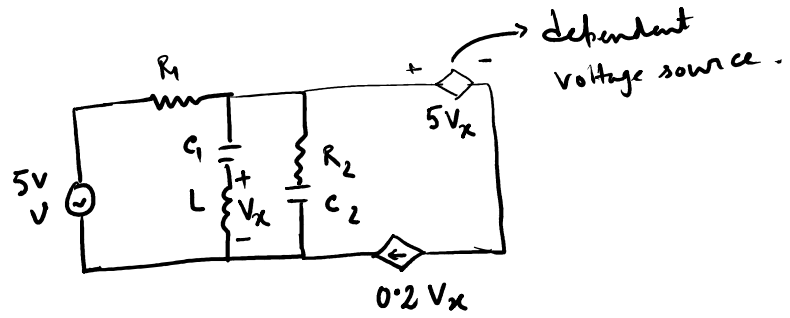
$$V = IR + \frac{1}{j\omega C} I$$

$$\int I e^{j\omega t} = \frac{I}{j\omega} e^{j\omega t} + C$$





Independent sources



dependent voltage source.